

System of Linear Equations

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Part I

Theory

Part II

Problem Set

Chapter 1

JEE Mains Problems

Q1 : Let $\alpha, \beta \in \mathbb{R}$ be such that the system of linear equations

$$x + 2y + z = 5$$

$$2x + y + \alpha z = 5$$

$$8x + 4y + \beta z = 18$$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

[JEE Mains 2026 2nd April Shift 1]

A) -4

B) 4

C) 8

D) -8

{ Solution : <https://www.youtube.com/watch?v=NAct7oeCy2s> }